Only tickets 1 and 3 are in the cup.



Problem 4: Add just one extra ticket to this cup: 2, 4, 5, or 6. Among red-showing rounds, which hidden color has more ways before and after each addition?

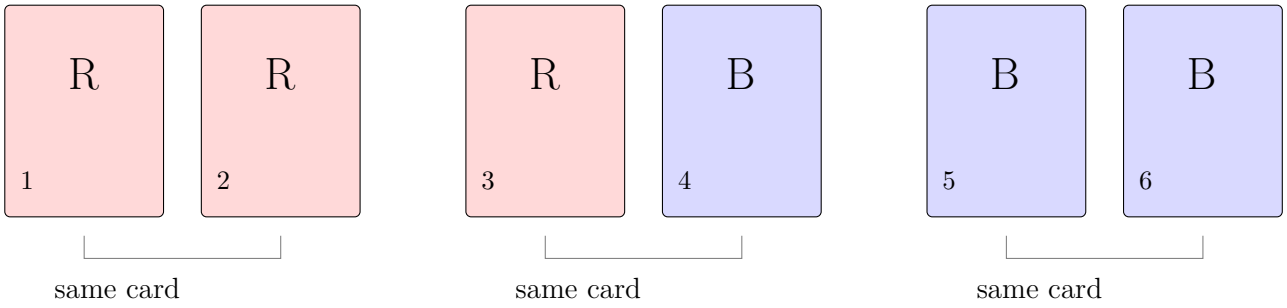
red underneath

blue underneath

Problem 5: Choose three tickets so red sometimes shows and then always hides blue. Find every choice that works.

Use all six face tickets again. Lay each card's two tickets on it, so removing a card takes both.

Problem 6: Remove one card so red showing always hides red. Can you remove a different card so red showing always hides blue?



Problem 7: Find every two-ticket cup where red showing gives a tie between the hidden colors. Red must sometimes show.